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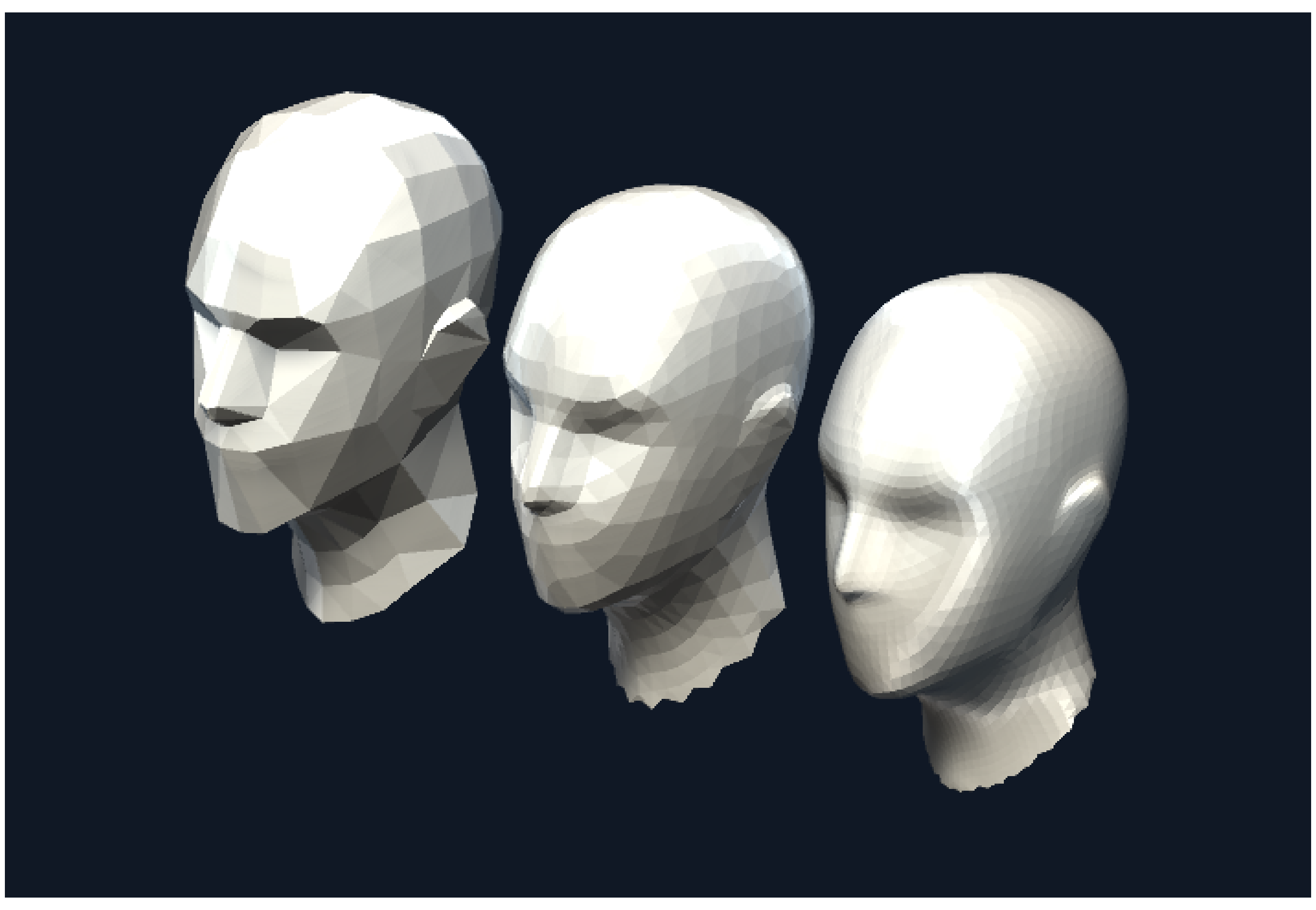
mattatz Add SubdivMorph demo. … 75858a3 on Jan 19, 2018 8 commits

Assets	Add SubdivMorph demo.	4 years ago
Captures	Add SubdivMorph demo.	4 years ago
ProjectSettings	Initial commit.	4 years ago
UnityPackageManager	Initial commit.	4 years ago
.gitignore	Initial commit.	4 years ago
LICENSE	Create LICENSE	4 years ago
README.md	Add SubdivMorph demo.	4 years ago

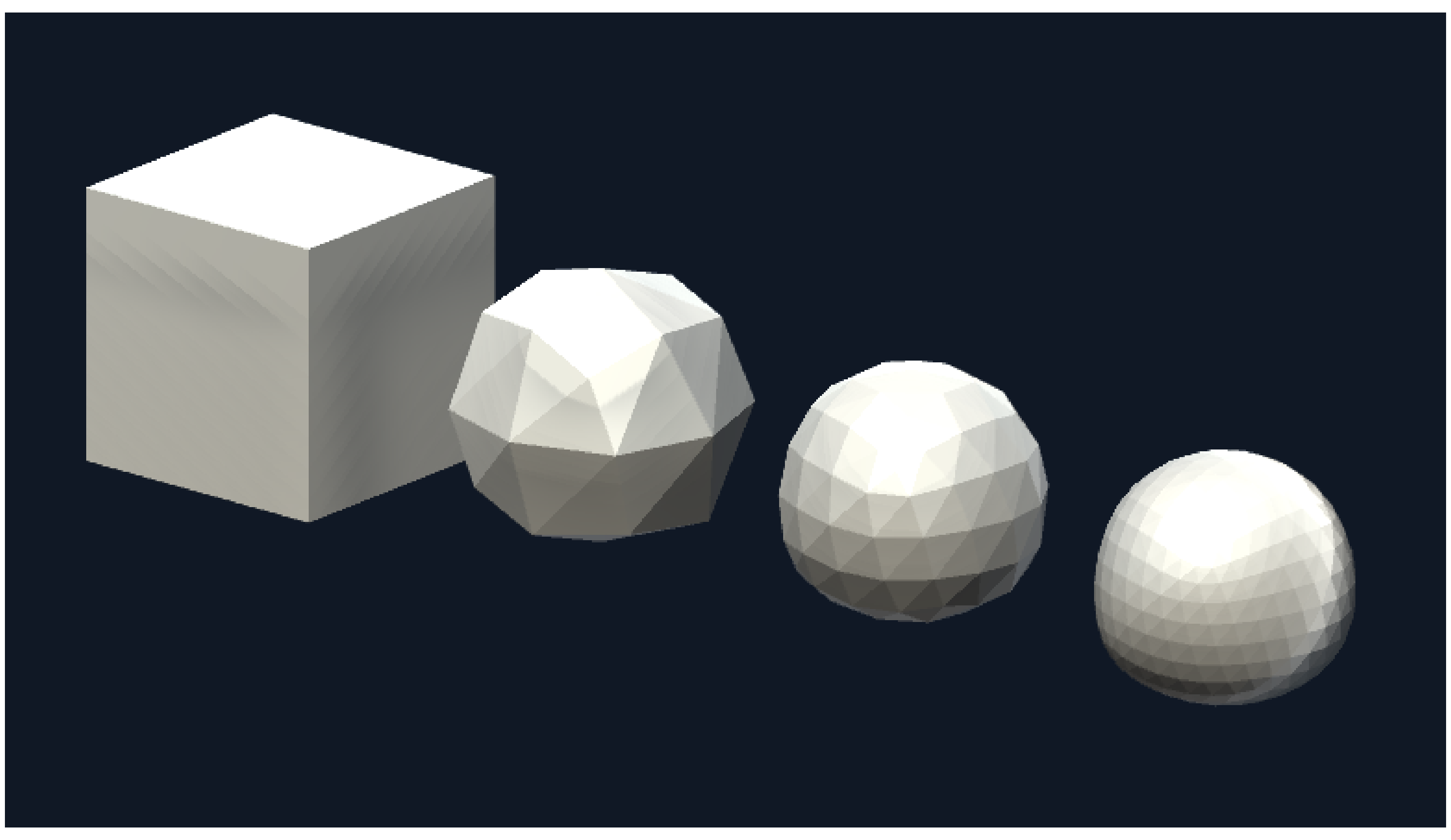
README.md

unity-subdivision-surface

Loop subdivision surface algorithm implementation in Unity.



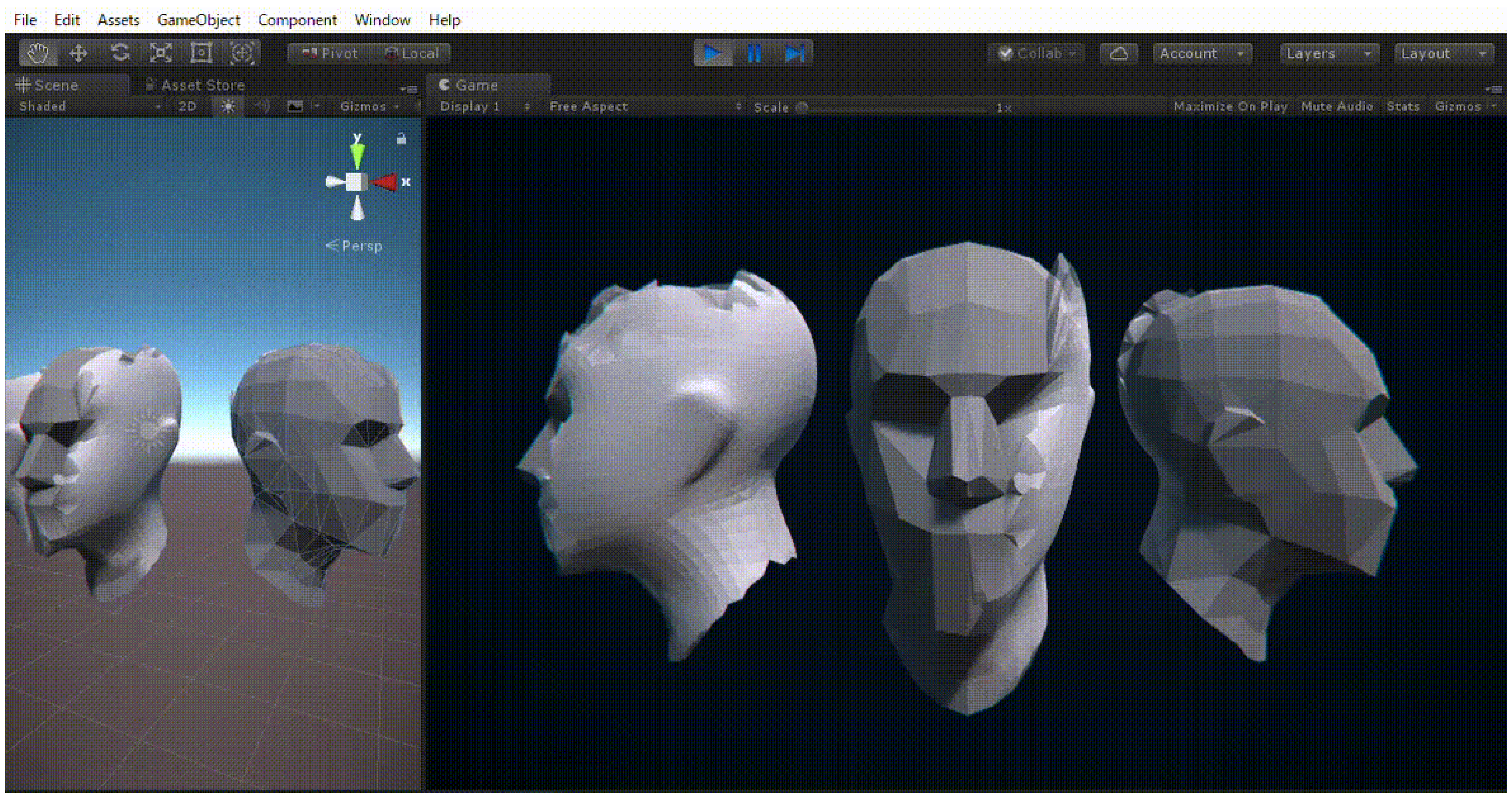
From left to right, original mesh, subdivided 1 time, subdivided 2times model.



Loop subdivision surface

> Loop subdivision surface is an approximating subdivision scheme developed by Charles Loop in 1987 for triangular meshes. wikipedia

SubdivMorph



SubdivMorph demo morph vertices between original and subdivided.

Usage

```
var filter = GetComponent<MeshFilter>();

// Require a mesh to weld (require to remove duplicated vertices)
var welded = SubdivisionSurface.Weld(filter.mesh, float.Epsilon, filter.mesh.bounds.size.x);

var mesh = SubdivisionSurface.Subdivide(
  welded,   // a welded mesh
  2,        // subdivision count
  false     // a result mesh is welded or not
);
filter.sharedMesh = mesh;
```

See demo scenes for details.

Compatibility

tested on Unity 2017.0.3, windows10 (GTX 1060).

Sources

- Free low poly head model by hexonian - https://sketchfab.com/models/988a1ffdb6244eaab9b293d296c6e868
- Carnegie Mellon Univ lecture slide - http://www.cs.cmu.edu/afs/cs/academic/class/15462-s14/www/lec_slides/Subdivision.pdf
- Michigan Tech Univ lecture slide - https://pages.mtu.edu/~shene/COURSES/cs3621/SLIDES/Subdivision.pdf
- Matt's Webcorner - Subdivision - https://graphics.stanford.edu/~mdfisher/subdivision.html

About

Loop subdivision surface algorithm implementation in Unity.

Readme

MIT License

Releases

No releases published

Packages

No packages published

Languages

- **C#** 55.7%
- **HLSL** 30.2%
- **ShaderLab** 14.1%

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